Agentic AI for Personalized Budget-Based Meal Planning

AD23731 Foundations of Agentic AI - Mini-Project

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BONAFIDE CERTIFICATE

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Faculty in-charge

Submitted to mini-project viva voce examination held on: ……

INTERNAL EXAMINER **EXTERNAL EXAMINER**

AGENTIC AI FOR PERSONALIZED BUDGET-BASED MEAL PLANNER

1. INTRODUCTION

1.1 Background and Motivation

Maintaining a healthy diet is an important part of achieving fitness and wellness goals. However, following a suitable diet is often difficult because nutritional requirements vary from person to person, while food prices and individual budgets also differ significantly.

Traditional diet plans generally provide predefined meals without sufficiently considering an individual's financial constraints, nutritional requirements, food preferences, fitness objectives, and day-to-day progress. A meal plan that is nutritionally appropriate may not always be affordable, while a low-cost diet may fail to provide sufficient calories or essential nutrients.

FoodIQ is proposed as an AI-powered personalized meal-planning platform that combines nutrition planning, budget management and Generative AI. The system generates personalized meal plans based on factors such as the user's budget, calorie requirements, fitness goal and dietary preference.

The platform is designed not simply as a static diet chart generator, but as an intelligent decision-support system. It processes user information, determines nutritional requirements, selects suitable food options under budget constraints, and uses Generative AI to provide recipes and conversational assistance.

FoodIQ therefore addresses two interconnected challenges:

1. **What should the user eat to satisfy nutritional requirements?**
2. **How can the required nutrition be achieved within the available budget?**

The combination of these constraints forms the central motivation behind FoodIQ.

1.2 Need for Intelligent and Agentic AI

Conventional meal-planning applications generally depend on predefined meal plans or simple recommendation rules. Such systems have limited ability to adapt recommendations according to multiple changing requirements.

FoodIQ introduces an AI-driven approach in which different tasks can be logically separated into specialized components.

For example:

- A nutrition component determines calorie requirements.
- A budget component checks affordability.
- A meal-planning component selects suitable foods.
- A recipe-generation component converts selected foods into practical recipes.
- A conversational AI component allows users to interact with the system.

An agentic extension of FoodIQ can coordinate these tasks through an orchestration layer. Instead of treating AI as a single response-generation component, the system can use task-oriented agents that exchange structured information and operate toward a common objective.

This approach is particularly useful because meal planning is a multi-step decision problem rather than a single prediction task.

1.3 Project Objectives

The major objectives of FoodIQ are:

- To generate personalized meal plans according to individual nutritional requirements.
- To consider the user's available food budget during meal planning.
- To calculate calorie requirements based on user characteristics and activity level.
- To support different fitness objectives such as weight loss, maintenance and muscle gain.
- To consider dietary preferences while selecting food.
- To provide practical meal and recipe recommendations using Generative AI.
- To provide nutrition tracking functionality.
- To provide an AI-based chatbot for user interaction.
- To create a scalable full-stack architecture for personalized nutrition management.
- To provide an intelligent framework that can be extended with real-time food-price information and knowledge retrieval.

2. PROBLEM STATEMENT AND BUSINESS CASE

2.1 Problem Definition

Many people want to maintain a healthy diet but face difficulties in creating a meal plan that simultaneously satisfies:

- nutritional requirements,
- fitness goals,
- personal food preferences,
- available budget,
- meal variety,
- preparation practicality.

Existing approaches commonly treat nutrition and affordability as separate problems.

A person may receive a recommended diet containing expensive ingredients that exceed their monthly budget. Alternatively, selecting only inexpensive foods without considering nutritional requirements can produce an unbalanced diet.

The problem addressed by FoodIQ can therefore be represented as:

Generate a personalized and nutritionally appropriate meal plan while considering the user's dietary preferences and financial constraints.

2.2 Real-World Context

Food expenditure is a recurring component of household expenses. Students, young professionals, gym users and budget-conscious individuals may have limited financial resources for food.

At the same time, nutritional requirements depend on several individual characteristics:

- Age
- Gender
- Height
- Weight
- Activity level
- Fitness goal
- Dietary preference

Consequently, a universal diet chart cannot effectively address every user's requirements.

FoodIQ attempts to bridge this gap by treating **nutrition and affordability as simultaneous constraints**.

2.3 Business Case

FoodIQ can be considered a digital nutrition-planning platform with potential applications in:

- Personal fitness
- Gym and wellness platforms
- Student nutrition
- Budget-conscious households
- Diet planning services
- Fitness coaching
- Nutrition applications

Potential future business models include:

- Freemium meal planning
- Premium personalized nutrition plans
- Fitness-center integration
- Nutritionist dashboards
- Grocery recommendations
- Food-product affiliate integration
- Subscription-based AI nutrition assistance

The current project focuses primarily on developing and demonstrating the technical platform rather than implementing a commercial business model.

3. USER AND STAKEHOLDER ANALYSIS

3.1 Target Users

FoodIQ primarily targets individuals who want personalized nutrition planning while maintaining control over their food expenditure.

Primary Users

Students

Students may have limited budgets while still wanting to maintain a healthy diet.

Gym and Fitness Users

Users with goals such as muscle gain, weight loss or maintenance can obtain plans based on their nutritional requirements.

Budget-Conscious Individuals

Users can specify the amount they are able to spend and obtain meal recommendations accordingly.

General Health-Conscious Users

Users interested in improving their eating habits can use the system for structured meal planning.

3.2 Stakeholders

The major stakeholders include:

Stakeholder	Role
End User	Provides personal information and follows the generated meal plan
Project Team	Develops and maintains FoodIQ
Nutrition Data Provider	Provides food and nutritional information
AI Service Provider	Provides Generative AI capabilities
Grocery/Product Platforms	Potentially provide product purchasing links
Fitness/Nutrition Professionals	Potential future domain stakeholders

3.3 User Requirements

Functional Requirements

The system should allow users to:

1. Create or enter their profile.
2. Provide age, gender, height and weight.
3. Select activity level.
4. Select fitness goal.

5. Enter food budget.
6. Specify dietary preference.
7. Generate personalized meal plans.
8. View nutritional information.
9. Generate recipes.
10. Track meals.
11. Interact with the AI chatbot.
12. Modify preferences.
13. Monitor progress.

Non-Functional Requirements

FoodIQ should provide:

- Usability
- Reliability
- Scalability
- Maintainability
- Responsive design
- Secure data handling
- Reasonable response time
- Consistent recommendations

4. PROPOSED AGENTIC AI SOLUTION

4.1 Overall Solution

FoodIQ combines a full-stack web application with AI-assisted nutrition planning.

The overall processing flow is:

User Input → User Profile Processing → Nutrition Requirement Calculation → Budget Analysis → Food Selection → Meal Plan Generation → AI Recipe Generation → Nutrition Tracking → User Feedback

The system can further be organized into specialized AI-oriented components.

4.2 Key Features

Personalized Meal Planning

The system generates meals based on the user's individual requirements instead of providing the same plan to every user.

Budget-Based Planning

Food selection considers the user's available budget.

Fitness Goal Support

The system can adapt recommendations according to:

- Weight Loss
- Weight Maintenance
- Muscle Gain

AI Recipe Generation

Google Gemini AI can be used to generate recipe instructions based on selected foods. Your project documentation confirms Gemini-based recipe generation and chatbot support.

Nutrition Tracking

Users can track their daily food consumption and progress.

AI Chatbot

The chatbot provides a conversational interface through which users can ask food and meal-related questions.

4.3 Agent-Based Approach

For an agentic implementation, FoodIQ can decompose the complete task into specialized responsibilities.

User Profile Agent

Processes:

- Age
- Gender
- Height
- Weight
- Activity level
- Goal
- Budget
- Dietary preference

Nutrition Agent

Determines the user's nutritional requirements.

Budget Agent

Evaluates food affordability and budget constraints.

Meal Planning Agent

Combines nutritional requirements and budget constraints to construct meals.

Recipe Agent

Uses Generative AI to create practical recipes.

Nutrition Assistant Agent

Handles user questions and provides contextual assistance.

Tracking Agent

Processes meal completion and progress information.

These components can be coordinated through a central orchestration layer.

5. DATASET / KNOWLEDGE SOURCES

5.1 Food Knowledge

FoodIQ requires structured information about food items, including:

- Food name
- Calories
- Protein
- Carbohydrates
- Fat
- Serving size
- Ingredients
- Estimated cost

The quality of the meal-planning system depends strongly on the quality and consistency of this information.

5.2 Nutrition Data

Nutrition information can be represented as structured records.

Example:

Food	Serving	Calories	Protein	Carbohydrates	Fat
Rice	100 g	Dataset value	Dataset value	Dataset value	Dataset value
Egg	1 unit	Dataset value	Dataset value	Dataset value	Dataset value
Dal	100 g	Dataset value	Dataset value	Dataset value	Dataset value

Important: actual numerical values should be populated from the dataset used by your implementation rather than invented values.

5.3 Price Information

The system can associate food items with estimated prices.

This enables the budget engine to calculate:

Total Meal Cost = Σ Cost of Ingredients

and:

Monthly Food Cost = Σ Daily Food Cost

The system can then compare estimated expenditure against the user's available budget.

5.4 Knowledge Base

The knowledge base can contain:

- Food nutrition
- Ingredients
- Recipes
- Dietary restrictions
- Meal combinations
- Cooking instructions
- Food substitutions

This knowledge can later be converted into a retrieval system for grounded AI responses.

6. TOOLS AND TECHNOLOGIES USED

The documented FoodIQ stack includes **HTML, CSS, JavaScript, Node.js, MongoDB Atlas and Google Gemini AI**.

Technology	**Purpose**
HTML5	Web page structure
CSS3	User interface styling
JavaScript	Client-side functionality
Node.js	Backend runtime
Express.js	Backend/API development where used
MongoDB Atlas	Database
Mongoose	MongoDB interaction where used
Google Gemini AI	Generative AI and chatbot
Chart.js	Nutrition/progress visualization where used
Git/GitHub	Source-code management

The exact technology list in the final report should match the final deployed implementation.

7. AGENT ARCHITECTURE

7.1 Architecture Overview

FoodIQ can be organized into the following logical layers:

Layer 1 — User Interface

Collects:

- Personal information
- Fitness goal
- Dietary preference
- Budget
- Meal preferences

Layer 2 — Application/API Layer

Processes user requests and coordinates system components.

Layer 3 — Intelligence Layer

Contains:

- Nutrition calculation
- Budget analysis
- Meal planning
- Recipe generation
- Conversational AI

Layer 4 — Knowledge/Data Layer

Contains:

- User profiles
- Food information
- Nutrition information
- Meal plans
- Tracking data
- Recipe information

Layer 5 — External AI Services

Google Gemini provides Generative AI capabilities

7.3 Tools

Possible tools associated with the agents include:

- Nutrition calculator
- Food database
- Budget calculator
- Recipe generator
- MongoDB queries
- Gemini API
- Nutrition tracker

7.4 Memory

FoodIQ can maintain user-specific information such as:

- Previous meal plans
- Food preferences
- Dietary restrictions
- Completed meals
- Previous questions
- Progress information

This allows future recommendations to be more contextual.

8. AGENT WORKFLOW AND ORCHESTRATION

8.1 Workflow Design

The workflow begins when the user submits their profile.

Step 1 — Input Collection

The system receives user information.

Step 2 — Profile Processing

The system converts the raw input into structured profile data.

Step 3 — Nutritional Calculation

The system determines estimated nutritional requirements.

Step 4 — Budget Processing

The system identifies the available spending constraint.

Step 5 — Food Selection

Food items are selected according to nutrition, dietary preferences and cost.

Step 6 — Meal Construction

Selected foods are organized into:

- Breakfast
- Lunch
- Dinner
- Snacks

Step 7 — Validation

The generated plan is checked against:

- Calories
- Protein
- Budget
- Dietary restrictions

Step 8 — Recipe Generation

The AI generates recipes and preparation instructions.

Step 9 — User Presentation

The final meal plan is presented through the web application.

9. AGENTIC RAG IMPLEMENTATION

This section should be included only to the extent that the final FoodIQ implementation actually contains retrieval functionality. Your documented project materials confirm RAG as a skill area, but the FoodIQ project description currently confirms Gemini recipe generation and chatbot support rather than a completed FoodIQ vector-RAG implementation.

For the report, the **proposed Agentic RAG architecture** can be described as follows.

9.1 Retrieval Architecture

Food/Nutrition Knowledge

↓

Preprocessing

↓

Embeddings

↓

Vector Database

↓

Retriever

↓

Relevant Food Context

↓

Gemini LLM

↓

Grounded Response

9.2 Knowledge Retrieval

When a user asks:

"Give me a high-protein breakfast within my budget."

The retrieval system can identify relevant:

- High-protein foods
- Breakfast recipes
- Food costs
- Nutritional information
- User dietary restrictions

The retrieved information is then supplied to the language model.

9.3 Generation Process

The LLM receives:

User Query + User Profile + Retrieved Food Knowledge + Budget Context

and produces a response based on this context.

This reduces dependence on the model's general knowledge for numerical nutrition and food-specific information.

9.4 Retrieval Quality Evaluation

Possible evaluation measures include:

- Retrieval relevance
- Context precision
- Context recall
- Answer groundedness
- Response relevance

These should be measured only if the retrieval system is actually implemented and tested.

10. SYSTEM IMPLEMENTATION

10.1 Frontend

The frontend provides interfaces for:

- User registration/profile
- Nutrition information
- Budget input
- Meal-plan generation
- Meal tracking
- Recipe viewing
- Chatbot interaction

HTML, CSS and JavaScript are documented as part of the FoodIQ implementation.

10.2 Backend

Node.js handles backend processing.

The backend is responsible for:

- Receiving user requests
- Processing profile information
- Performing calculations
- Communicating with MongoDB
- Calling Gemini AI
- Generating responses
- Returning data to the frontend

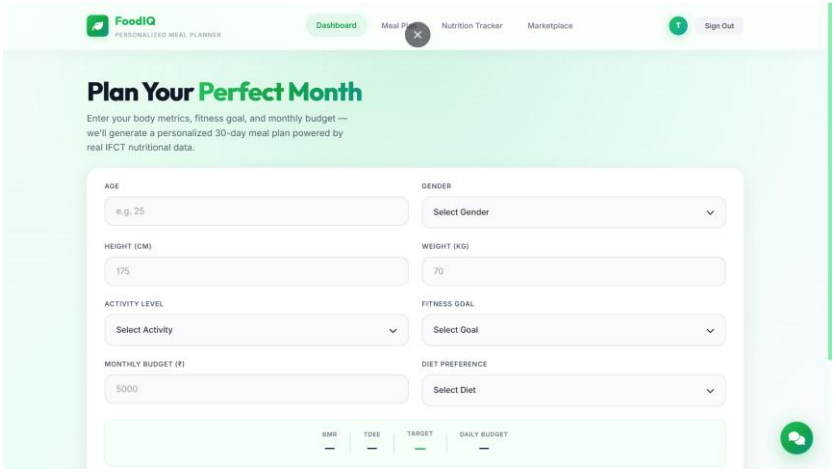
10.4 AI Integration

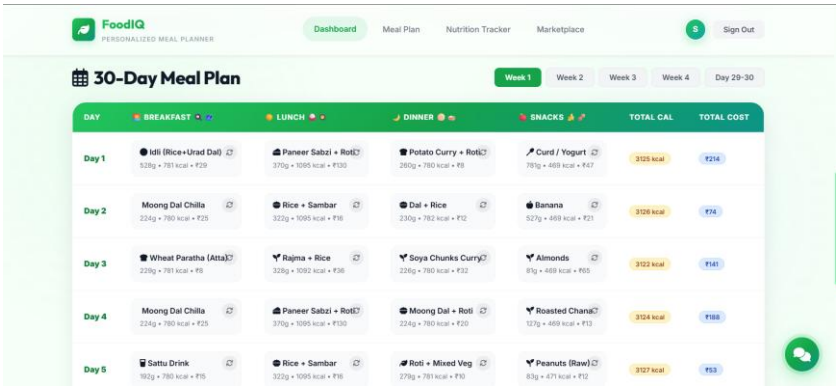
Google Gemini AI is integrated for:

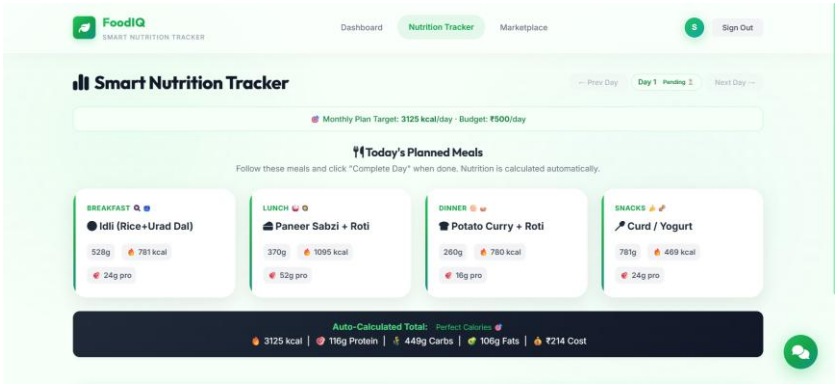
- Recipe generation
- Conversational chatbot
- Food-related responses
- Personalized explanations

Your project documentation specifically identifies Gemini AI for AI-powered recipe generation and chatbot functionality.

10.5 Prototype Screenshots







11. EXPERIMENTAL EVALUATION AND RESULTS

11.1 Evaluation Strategy

FoodIQ should be evaluated based on whether it successfully satisfies the user's combined nutrition and budget requirements.

Important evaluation dimensions include:

- Nutritional adequacy
- Budget compliance
- Recommendation relevance

- Response time
- Recipe quality
- User satisfaction

11.2 Budget Compliance

For every generated meal plan:

Budget Compliance = Generated Food Cost ≤ User Budget

The system should identify plans that exceed the specified budget.

11.3 Nutritional Accuracy

The generated plan can be compared against the user's calculated nutritional requirement.

For example:

Required Calories

↓

Generated Plan Calories

↓

Calculate Difference

The same process can be applied to protein and other nutrients.

11.4 Recommendation Relevance

Recommendations can be evaluated according to:

- Goal compatibility
- Dietary compatibility
- Budget compatibility
- Nutritional suitability

11.5 AI Response Evaluation

The chatbot and recipe generation can be evaluated using:

- Relevance
- Completeness
- Groundedness
- Response consistency
- User feedback

11.6 Performance Evaluation

System-level measurements can include:

Metric	Purpose
API Response Time	Backend performance
Meal Generation Time	Planning efficiency
AI Response Time	Gemini integration performance
Database Query Time	Data-access efficiency
Error Rate	Reliability

Actual numerical values should be inserted from your testing rather than fabricated.

12. COMPARISON, INNOVATION AND SCALABILITY

12.1 Conventional Meal Planning vs FoodIQ

Aspect	Conventional Diet Plan	FoodIQ
Personalization	Limited	User-specific
Budget consideration	Often separate	Integrated
Fitness goal	Fixed/custom manually	Included in planning
AI assistance	Usually absent	Gemini-powered
Recipe generation	Predefined	AI-assisted
Tracking	May be separate	Integrated
Conversational support	Usually absent	Chatbot
Dynamic extension	Limited	Designed for extension

This comparison describes system characteristics rather than assigning an overall ranking.

12.2 Innovation

FoodIQ combines multiple requirements that are commonly handled separately:

Nutrition + Budget + Fitness Goal + Dietary Preference + Generative AI + Tracking

The central innovation is therefore the integration of **financial constraints into personalized nutrition planning**.

12.3 Scalability

FoodIQ can be extended from an individual meal planner into a broader nutrition platform.

Database Scalability

MongoDB Atlas can support increasing user and meal-plan data.

AI Scalability

The AI layer can be extended with more specialized capabilities.

Knowledge Scalability

Additional food, recipe and nutrition information can be added.

Application Scalability

The backend can eventually support:

- Mobile applications
- Multiple users
- Fitness-center integration
- Nutritionist dashboards

12.4 Real-World Applicability

Potential applications include:

- College students
- Gym users
- Working professionals
- Budget-conscious families
- Fitness coaches
- Nutrition applications

A future version could connect meal recommendations with grocery platforms, enabling users to move from **meal planning → ingredient selection → purchasing**.

13. CONCLUSION AND FUTURE WORK

13.1 Conclusion

FoodIQ is an AI-powered personalized meal-planning platform designed to address the practical challenge of maintaining a suitable diet under financial constraints.

Unlike a static diet chart, the system considers individual characteristics and preferences when generating recommendations. The platform combines nutrition planning with budget awareness and Generative AI capabilities.

The full-stack architecture consists of a web-based frontend, Node.js backend, MongoDB Atlas database and Google Gemini AI integration. The documented implementation provides personalized meal planning, AI-powered recipe generation, nutrition tracking and chatbot support.

The proposed agent-oriented architecture further provides a framework for decomposing the planning process into specialized tasks such as profile analysis, nutrition calculation, budget analysis, meal planning, recipe generation and tracking.

The project demonstrates how AI can be integrated into a practical application where the objective is not merely to generate text, but to support a multi-constraint decision-making process.

13.2 Future Work

1. Real-Time Food Price Tracking

Food prices can change over time. Future versions can integrate real-time pricing information to dynamically update meal plans.

2. Advanced Agentic Architecture

Specialized agents can be implemented using an orchestration framework to improve task coordination.

3. Agentic RAG

A vector-based knowledge retrieval layer can provide grounded food and nutrition information to the AI system.